VANSH TOMAR

+91 7999826982 | Madhya Pradesh, India
vansh.tomar.0306@gmail.com | LinkedIn | GitHub | Portfolio

ABOUT ME

Mathematic and Computing student specializing in Artificial Intelligence and Machine Learning, with hands-on experience building machine learning, generative AI, agentic AI, and full-stack applications. Experienced in Python, data processing, REST APIs, and geospatial data, with a focus on developing practical AI-driven solutions.

EDUCATION

Madhav Institute of Technology and Science — Gwalior, India
BTech, MAC (CGPA: 9.17) — *Aug 2025 – Expected Aug 2029*

Swami Vivekananda Academy — Gwalior, India
CBSE Class 12th PCM, Percentage: ***91.8%*** — *Apr 2023 – Jun 2024*

PROJECTS

AI Shopping Agent — *Python, FastAPI, LLMs, Groq API*

- Developed an agentic AI shopping assistant capable of interpreting user requirements and assisting with product discovery and comparison.
- Integrated LLM and vision capabilities through APIs to process product information and generate context-aware recommendations.
- Built a Python-based backend workflow for coordinating AI reasoning, product data processing, and user queries.

Neural Network from Scratch — *Python, NumPy, Machine Learning, Deep Learning*

- Implemented a neural network from scratch in Python and NumPy to understand the mathematical foundations of forward propagation and backpropagation.
- Implemented core components including weighted layers, activation functions, loss computation, gradient calculation, and parameter updates.
- Trained and evaluated the model while analyzing learning behavior and model performance without relying on high-level deep learning frameworks.

Urban Heat Mitigation using AI/ML — *Python, Machine Learning, Google Earth Engine*

- Developed an AI/ML approach for analyzing urban heat patterns in Delhi using satellite-derived environmental and urban features.
- Processed Landsat and Sentinel-2 data to derive Land Surface Temperature (LST), NDVI, NDWI, EVI, NDBI, land cover, tree cover, elevation, slope, and distance-to-water features.
- Built a machine learning pipeline for predicting urban heat conditions and evaluating potential cooling interventions such as vegetation, cool roofs, increased albedo, and water bodies.

SKILLS

Programming: Python, C++, JavaScript, SQL

AI/ML: Machine Learning, Deep Learning, Neural Networks, Computer Vision, Large Language Models (LLMs), Generative AI, Agentic AI, Time-Series Analysis

Data Science: NumPy, Pandas, Scikit-learn, xarray, Dask, Matplotlib

Backend: FastAPI, REST APIs, Groq API

Frontend: React, JavaScript, Leaflet

Databases & Authentication: MongoDB Atlas, Firebase, Google OAuth

Geospatial & Remote Sensing: Google Earth Engine, Landsat, Sentinel-2, GLORYS, OSTIA, ERA5-Land, NetCDF

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, LaTeX